14 Path Integral

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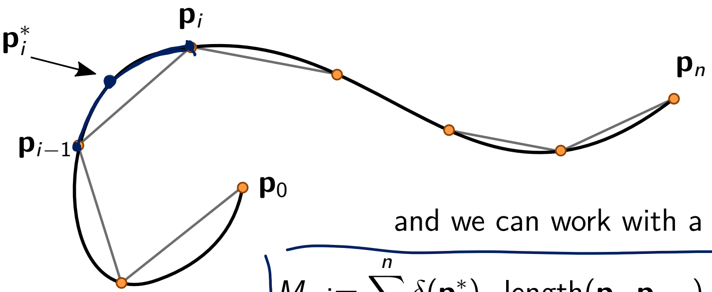
Motivation

Suppose we'd like to evaluate the total mass of a weighted wire with known density δ.

We can choose the points $\mathbf{p}_0, \mathbf{p}_1, \ldots, \mathbf{p}_n$ on the wire, splitting the wire into segments. The mass of each segment is approximately

$$m_i \approx \delta(\mathbf{p}_i^*) \cdot \text{length}(\mathbf{p}_i, \mathbf{p}_{i-1}),$$



and we can work with a Riemann sum, for instance,

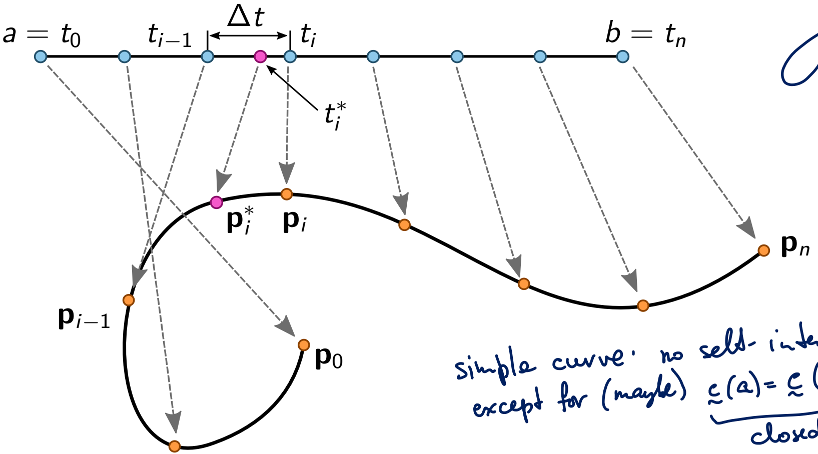
$$\boxed{M_n := \sum_{i=1}^{n} \delta(\mathbf{p}_i^*) \cdot \text{length}(\mathbf{p}_i, \mathbf{p}_{i-1}),}$$

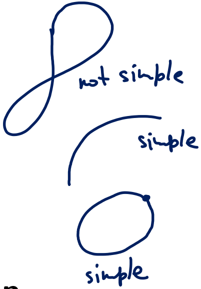
to define a path integral.

Integration over a path

not self-intersecting

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $\mathbf{c} : [a, b] \to \mathbb{R}^2$ be a simple path. We would like to work on the domain of the path instead of its image (trace).





simple curve: no self-intersection, except for (maybe) $\underset{\sim}{c}(a) = \underset{\sim}{c}(b)$

closed

Definition 14.1 (Path Integral)

Let $\mathbf{c} : [a, b] \to \mathbb{R}^n$ be a simple C^1 path, and let $f : \mathbb{R}^n \to \mathbb{R}$ be such that the composite function $t \mapsto f(\mathbf{c}(t))$ is continuous on $[a, b]$. Then the path integral of the function f over the curve $C = \mathbf{c}([a, b])$ is

$$\int_C f\, ds = \boxed{\int_a^b f(\mathbf{c}(t))\|\mathbf{c}'(t)\|\, dt.}$$

When $f(\mathbf{x}) = 1$, this integral gives a familiar expression for the length of a path/curve.

$$\int_a^b \|\underset{\sim}{c}'(t)\|\, dt$$

Example 14.2

Evaluate the integral

$f(x,y,z) = xyz$

$\underset{\sim}{c}(t) = \begin{pmatrix} t \\ \frac{1}{3} 2\sqrt{2}\, t^{3/2} \\ \frac{1}{2} t^2 \end{pmatrix}$

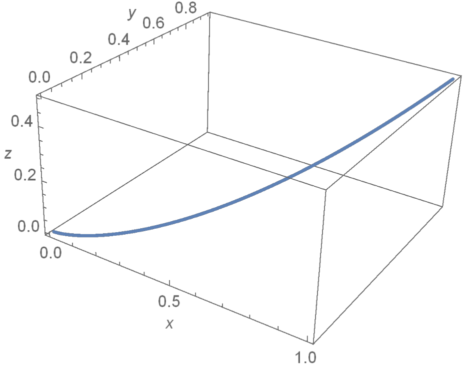
$$\int_C xyz \, ds,$$

where C is the curve defined as $x(t) = t$, $y(t) = \frac{1}{3}\sqrt{8t^3}$, $z(t) = \frac{1}{2}t^2$, $0 \le t \le 1$.

$$\int_C xyz \, ds = \int_0^1 f(\underset{\sim}{c}(t)) \cdot \|\underset{\sim}{c}'(t)\| \, dt =$$

$$\underset{\sim}{c}'(t) = \begin{pmatrix} 1 \\ \sqrt{2}\, t^{1/2} \\ t \end{pmatrix} \qquad \|\underset{\sim}{c}'(t)\| = \sqrt{\underbrace{1 + 2t + t^2}_{(1+t)^2}} = (1+t)$$

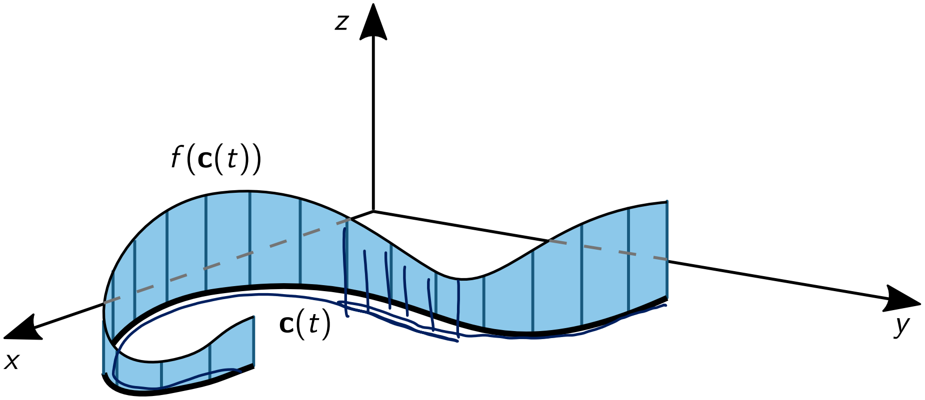
$$f(\underset{\sim}{c}(t)) = t \cdot \frac{1}{3} \cdot \cancel{2}\sqrt{2}\, t^{3/2} \cdot \frac{1}{\cancel{2}} t^2 = \frac{\sqrt{2}}{3} t^{9/2}$$



$$= \int_0^1 \frac{\sqrt{2}}{3} t^{9/2}(1+t)\,dt = \dots = \boxed{\frac{16\sqrt{2}}{143}}$$

Why do we care that the curve is simple?

An alternative interpretation of the path integral is the area under the graph of a function defined on a curve. If the parametrisation is chosen incorrectly, for instance, two segments of a curve overlap, we may obtain an incorrect value when evaluating a path integral.



Example 14.3 (Inappropriate parametrisation)

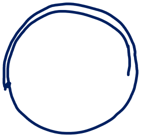
We evaluate the circumference of the unit circle using a path integral.

Correct calculation:

$$\mathbf{c}(t) = (\cos t, \sin t),\ 0 \le t \le 2\pi,$$

$$\int_C 1\ ds = \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t}\ dt = \int_0^{2\pi} dt = 2\pi.$$

Incorrect calculation: choosing a parametrisation with an overlap,



$$\mathbf{c}(t) = (\cos t, \sin t),\ 0 \le t \le 3\pi,$$

we obtain an absurd answer

$$\int_C 1\ ds \quad \text{“}=\text{”} \quad \int_0^{3\pi} \sqrt{\sin^2 t + \cos^2 t}\ dt = \int_0^{3\pi} dt = 3\pi.$$

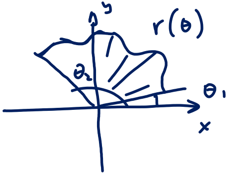
Example 14.4

When the curve is described via its equation in polar coordinates as $\boxed{r = r(\theta)}$, show that the integral of a function f over this curve can be evaluated using the formula

$$\int_C f(x, y)\, ds = \int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r^2 + r'^2}\, d\theta.$$

Note that for $f(x, y) = 1$ we obtain an expression for the length of a curve in polar coordinates,

$$\int_C ds = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + r'^2}\, d\theta.$$



$r = r(\theta) \qquad \underset{\sim}{c}(\theta) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r(\theta)\cos\theta \\ r(\theta)\sin\theta \end{pmatrix}$

$$\underset{\sim}{c}'(\theta) = \begin{pmatrix} -r(\theta)\sin\theta + r'(\theta)\cos\theta \\ r(\theta)\cos\theta + r'(\theta)\sin\theta \end{pmatrix} \qquad \|\underset{\sim}{c}'(\theta)\| = \sqrt{\left(-r(\theta)\sin\theta + r'(\theta)\cos\theta\right)^2 + \left(r(\theta)\cos\theta + r'(\theta)\sin\theta\right)^2}$$

$$= \sqrt{r^2(\theta)\underbrace{\sin^2\theta} - \cancel{2rr'\sin\theta\cos\theta} + (r'(\theta))^2\underbrace{\cos^2\theta} + r^2(\theta)\underbrace{\cos^2\theta} + \cancel{2rr'\cos\theta\sin\theta} + (r'(\theta))^2\underbrace{\sin^2\theta}}$$

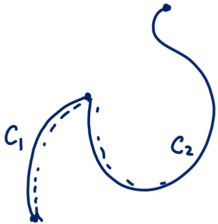
$$= \sqrt{r^2(\theta) + (r'(\theta))^2}$$

$$\int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r^2 + (r')^2}\, d\theta$$

$r(\theta)$

Properties of the path integral

- $$\int_C f_1\, ds + \int_C f_2\, ds = \int_C (f_1 + f_2)\, ds$$ – given that $\int_C f_1\, ds$ & $\int_C f_2\, ds$ are well-defined
- $$\int_C k\ f\ ds = k \int_C f\ ds$$ for constant $k \in \mathbb{R}$
- $$\int_C f\ ds = \int_{C_1} f\ ds + \int_{C_2} f\ ds,$$ where C consists of two non-overlapping segments C_1 and C_2, sometimes written as $C = C_1 + C_2$.

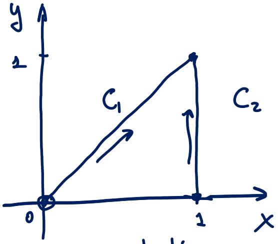


Path integral doesn't depend on the orientation of the curve!

Example 14.5 (Additivity)

Let C be a piecewise linear curve that consists of two line segments: from $(0,0)$ to $(1,1)$ and from $(1,1)$ to $(1,0)$. Evaluate

$$I = \int_C (x + y^2)\, ds.$$



C_1: $\underset{\sim}{c}_1(t) = \begin{pmatrix} t \\ t \end{pmatrix}, \quad t \in [0,1]$

A, B

$(1-t)A + tB, \; t \in [0,1]$

$\begin{pmatrix} 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} t \\ t \end{pmatrix}$

C_2: $\underset{\sim}{c}_2(t) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ t \end{pmatrix}, \quad t \in [0,1]$

orientation doesn't matter,

$\|\underset{\sim}{c}_1'(t)\| = \|(1,1)\| = \sqrt{2}, \qquad \|\underset{\sim}{c}_2'(t)\| = \|(0,1)\| = \underline{1}$

$I = I_1 + I_2, \qquad I_1 = \int_{C_1} \underline{(x+y^2)}\, ds, \qquad I_2 = \int_{C_2} (x+y^2)\, ds$

$$I_1 = \int_0^1 \underbrace{(t+t^2)}_{f(\underset{\sim}{c}(t))} \cdot \underbrace{\sqrt{2}}_{\|\underset{\sim}{c}'(t)\|} \, dt = \quad . \quad = \frac{5\sqrt{2}}{6}$$

$$\int_a^b f(\underset{\sim}{c}(t)) \, \underline{\|\underset{\sim}{c}'(t)\|} \, dt$$

$a < b$

$$I_2 = \int_0^1 (1+t^2) \cdot 1 \, dt = \quad \ldots \quad = \frac{4}{3}$$

$$\therefore \; I = I_1 + I_2 = \frac{5\sqrt{2}}{6} + \frac{4}{3}$$

$$\boxed{\int_1^0 (1+t^2) \cdot 1 \, dt = -\frac{4}{3}}$$

Basic applications

- Mass of a wire: $M = \int_C \delta(x,y)\,ds$, here $\delta(x,y)$ is the density.
- Centre of mass: $\bar{x} = \frac{1}{M}\int_C x\delta(x,y)\,ds$, $\quad \bar{y} = \frac{1}{M}\int_C y\delta(x,y)\,ds$.

Analogous expressions for the three-dimensional case.

Example 14.6

A wire of density $\delta(x,y,z) = 15\sqrt{y+2}$ lies along the curve C parametrised by

$$\mathbf{c}(t) = (t^2-1)\mathbf{j} + 2t\mathbf{k}, \quad 0 \le t \le 1.$$

Find the centre of mass of this wire.

$\underset{\sim}{c}'(t) = \begin{pmatrix} 0 \\ 2t \\ 2 \end{pmatrix}$ $\quad \|\underset{\sim}{c}'(t)\| = \sqrt{4t^2+4} = 2\sqrt{t^2+1}$

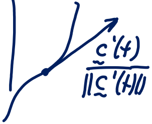
$$M = \int_C \delta \, ds = \int_0^1 \delta(\underset{\sim}{c}(t)) \cdot \|\underset{\sim}{c}'(t)\| \, dt = 15\int_0^1 \sqrt{\underbrace{t^2-1+2}_{\sqrt{t^2+1}}} \; 2\sqrt{t^2+1} \, dt = \boxed{40}$$

$$\bar{x} = 0 \, , \quad \bar{y} = \frac{-24}{40} \, , \quad \bar{z} = \frac{45}{40}$$

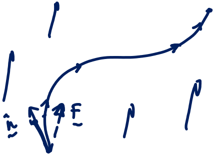
Vector path integrals

To calculate the work of a force field $\mathbf{F}$ on an object that is moved along an oriented path C (defined by $\mathbf{c} : [a, b] \to \mathbb{R}$), we integrate the projection of this force onto the unit tangent vector of the curve:

$$\int_a^b \underbrace{\left(\mathbf{F}(\mathbf{c}(t)) \cdot \frac{\mathbf{c}'(t)}{\|\mathbf{c}'(t)\|}\right)}_{f(\mathbf{c}(t))} \|\mathbf{c}'(t)\|\, dt = \boxed{\int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt.}$$



This is called the vector path integral of $\mathbf{F}$ over an oriented curve C and is denoted by



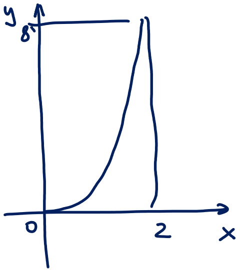
$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt.$$

Different parametrisations of the same oriented curve give the same value of this integral, however changing the orientation of the curve changes the sign of the integral.

Example 14.7

Find a parametrisation of the curve given by $y = x^3$ between $(0,0)$ and $(2,8)$. Calculate $\int_C \mathbf{F} \cdot d\mathbf{s}$ for

$$\mathbf{F} = (2xy^3 - 3x^2)\mathbf{i} + (3x^2y^2 + 2)\mathbf{j}.$$



$$\underset{\sim}{c}(t) = \begin{pmatrix} t \\ t^3 \end{pmatrix}, \quad t \in [0,2]$$

$$\underset{\sim}{c}'(t) = \begin{pmatrix} 1 \\ 3t^2 \end{pmatrix}$$

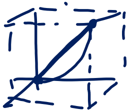
$$\underset{\sim}{F}(\underset{\sim}{c}(t)) = \begin{pmatrix} 2t \cdot t^9 - 3t^2 \\ 3t^2 \cdot t^6 + 2 \end{pmatrix} = \begin{pmatrix} 2t^{10} - 3t^2 \\ 3t^8 + 2 \end{pmatrix}$$

$$I = \int_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \int_0^2 \begin{pmatrix} 2t^{10} - 3t^2 \\ 3t^8 + 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3t^2 \end{pmatrix} dt = \int_0^2 2t^{10} - 3t^2 + 9t^{10} + 6t^2 \, dt = \dots = \underline{2056}$$

Example 14.8

Let $\mathbf{F}(x, y, z) = (y - x^2, z - y^2, x - z^2)$, and let C be the curve from $(0, 0, 0)$ to $(1, 1, 1)$ parametrised by

$$\mathbf{c}(t) = (t, t^2, t^3),\ 0 \le t \le 1.$$



a. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$;

$s = t^2$

$s \quad s^2 \quad s^3$

b. evaluate the same integral using a different parametrisation $\mathbf{r}(t) = (t^2, t^4, t^6)$, $0 \le t \le 1$ of the same curve;

c. evaluate the integral $\int_C \mathbf{F} \cdot d\mathbf{s}$ along the straight line connecting the points $(0, 0, 0)$ and $(1, 1, 1)$.

a. $\displaystyle\int_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \int_0^1 \begin{pmatrix} \overset{=0}{t^2 - t^2} \\ t^3 - t^4 \\ t - t^6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix} dt = \int_0^1 2t^4 - 2t^5 + 3t^3 - 3t^8 \, dt = \dots = \frac{29}{60}$

b. $$\int_C \underset{\sim}{F} \, d\underset{\sim}{s} = \int_0^1 \begin{pmatrix} t^4 - t^4 \\ t^6 - t^3 \\ t^2 - t^{12} \end{pmatrix} \cdot \begin{pmatrix} 2t \\ 4t^3 \\ 6t^5 \end{pmatrix} dt = \dots\dots = \frac{29}{60}$$

c. $$\underset{\sim}{c}(t) = \begin{pmatrix} t \\ t \\ t \end{pmatrix}$$

$$\int_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \int_0^1 \begin{pmatrix} t - t^2 \\ t - t^2 \\ t - t^2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} dt = \dots = \frac{1}{2}$$

A different piece of notation for vector line integrals

Using the notation

$$\mathbf{c}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \quad \text{and} \quad \mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k},$$

we can write

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt$$

$$= \int_a^b \left[F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right] dt$$

$$= \int_C F_1\, dx + F_2\, dy + F_3\, dz.$$

This is **alternative notation** for vector line integrals.

Example 14.9

Evaluate

$$I = \int_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \int_C \underbrace{2y}_{F_1}\,dx \underbrace{- 3}_{F_2}\,dy,$$

Recognising the notation!

$$\underset{\sim}{F} = \begin{pmatrix} 2y \\ -3 \end{pmatrix}$$

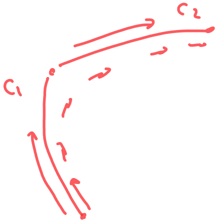
where C is the curve parametrised by $\mathbf{c}(t) = (t^2, t^3)$, $0 \leq t \leq 2$.

$$I = \int_0^2 \begin{pmatrix} 2t^3 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2t \\ 3t^2 \end{pmatrix} dt = \int_0^2 4t^4 - 9t^2 \, dt = \ldots = \frac{8}{5}$$

Properties of the vector line integral

Here C is a piecewise smooth oriented curve, $\mathbf{F}$ and $\mathbf{G}$ are vector fields.

- $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{s} + \int_C \mathbf{G}\cdot d\mathbf{s} = \int_C (\mathbf{F}+\mathbf{G})\cdot d\mathbf{s}$
- $\displaystyle\int_C k\,\mathbf{F}\cdot d\mathbf{s} = k\int_C \mathbf{F}\cdot d\mathbf{s}$ for constant $k\in\mathbb{R}$
- $\displaystyle\int_{-C} \mathbf{F}\cdot d\mathbf{s} = -\int_C \mathbf{F}\cdot d\mathbf{s}$, where $-C$ is C reversed.
- $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{s} = \int_{C_1} \mathbf{F}\cdot d\mathbf{s} + \int_{C_2} \mathbf{F}\cdot d\mathbf{s}$, where C consists of two non-overlapping (oriented) segments C_1 and C_2.



The curve C needs to be **oriented**: the sign of the integral changes if the curve is reversed.

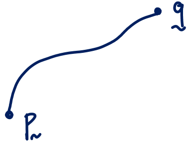
Theorem 14.10 (Fundamental for conservative vector fields)

If $\mathbf{F} = \nabla\phi$ *on some open set* D*, then for every oriented curve* C *in* D *with an initial point* $\mathbf{p}$ *and terminal point* $\mathbf{q}$

$\underset{\sim}{c} : [a, b] \to \mathbb{R}^n$

$\underset{\sim}{c}(a) = \underset{\sim}{p}$

$\underset{\sim}{c}(b) = \underset{\sim}{q}$

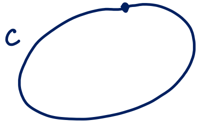


$$\int_C \mathbf{F} \cdot ds = \phi(\mathbf{q}) - \phi(\mathbf{p}).$$

$$\int_a^b \underbrace{\underset{\sim}{F}(\underset{\sim}{c}(t))}_{\nabla\varphi(\underset{\sim}{c}(t))} \cdot \underset{\sim}{c}'(t)\, dt = \int_a^b \underbrace{\nabla\varphi(\underset{\sim}{c}(t)) \cdot \underset{\sim}{c}'(t)}_{(\varphi(\underset{\sim}{c}(t)))'_t}\, dt$$

$$= \varphi(\underbrace{\underset{\sim}{c}(b)}_{\underset{\sim}{q}}) - \varphi(\underbrace{\underset{\sim}{c}(a)}_{\underset{\sim}{p}})$$

If the curve C is closed, then



$$\int_C \mathbf{F} \cdot ds = \oint_C \mathbf{F} \cdot ds = 0.$$

the curve C is closed

path independent everywhere $\Leftrightarrow$ integral vanishes on loops everywhere $\Leftrightarrow$ $\mathbf{F} = \nabla\phi$

A physical interpretation of this result is that the work done by a conservative force on an object that moves between two points in space does not depend on the path.

Example 14.11

Let $\phi = xy + xz^2$ and let $\mathbf{F} = \nabla\phi$, that is, $\mathbf{F}$ is conservative with potential ϕ.

a. Integrate $\mathbf{F}$ along the curve C_1: $\mathbf{c}_1(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + \frac{2t}{\pi}\mathbf{k}$ from $(1, 0, 0)$ to $(0, 1, 1)$.

b. Integrate $\mathbf{F}$ along the curve C_2: $\mathbf{c}_2(t) = (1-t)\mathbf{i} + t\mathbf{j} + t\mathbf{k}$ from $(1, 0, 0)$ to $(0, 1, 1)$.

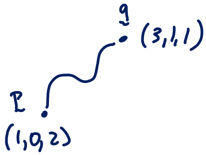
a $$\int_0^1 \underset{\sim}{F}(\underset{\sim}{c}(t))\cdot \underset{\sim}{c}'(t)\,dt = \phi(\underbrace{\underset{\sim}{c}(b)}_{(0,1,1)}) - \phi(\underbrace{\underset{\sim}{c}(a)}_{(1,0,0)}) = 0 - 0 = \underline{0}$$

b. The same! $$\int_0^1 \underset{\sim}{F}\cdot d\underset{\sim}{s} = \phi(\underset{\sim}{c}(b)) - \phi(\underset{\sim}{c}(a)) = \underline{0}$$

Example 14.12

Find the potential function for

$\underset{\sim}{F} = \nabla \varphi$



$$\mathbf{F}(x, y, z) = (yz^2 + 1)\mathbf{i} + (xz^2 - ze^y)\mathbf{j} + (2xyz - e^y)\mathbf{k},$$

and hence calculate the work done by a force $\mathbf{F}$ on an object as it moves from $(1, 0, 2)$ to $(3, 1, 1)$.

$$\frac{\partial \varphi}{\partial x} = yz^2 + 1$$

$$\frac{\partial \varphi}{\partial y} = xz^2 - ze^y$$

$$\frac{\partial \varphi}{\partial z} = 2xyz - e^y$$

$$\varphi(x,y,z) = xyz^2 + x + f_1(y,z)$$

$$\varphi(x,y,z) = xyz^2 - ze^y + f_2(x,y)$$

$$\varphi(x,y,z) = xyz^2 - ze^y + f_2(x,y)$$

$\underset{\sim}{F} = \nabla \varphi,$

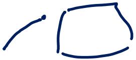
$\varphi = xyz^2 - ze^y + x$

$$\int_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \varphi(\underset{\sim}{q}) - \varphi(\underset{\sim}{p})$$

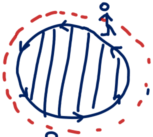
$$= (3 \cdot 1 \cdot 1 - e + 3) - (0 - 2 \cdot 1 + 1)$$

$$= 6 - e + 2 - 1 = \boxed{7 - e}$$

Theorem 14.13 (Green)

 $\cancel{dx}$ 

Let C be a positively oriented piecewise smooth simple closed curve in a plane, let Ω be the bounded region enclosed by C, and let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j}$ be a vector field. If F_1, F_2, $\frac{\partial F_2}{\partial x}$ and $\frac{\partial F_1}{\partial y}$ are defined and continuous in an open region containing Ω, then



$$\oint_C F_1\,dx + F_2\,dy = \iint_\Omega \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \underbrace{dA}_{dx\,dy}.$$

$$\underset{\sim}{F} = (F_1, F_2) = (F_1, 0) + (0, F_2)$$

$$\oint_C \underset{\sim}{F}\cdot d\underset{\sim}{s} = \overbrace{\oint_C (F_1, 0)\cdot d\underset{\sim}{s}}^{I} + \oint_C (0, F_2)\cdot d\underset{\sim}{s}$$

$$\iint_\Omega \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA = \iint_\Omega \frac{\partial F_2}{\partial x}\, dA \overbrace{- \iint_\Omega \frac{\partial F_1}{\partial y}\, dA}^{J}$$

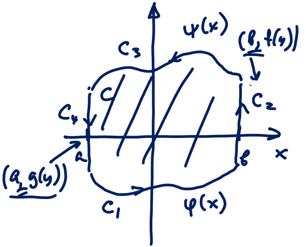
To show:
$I = J$
(similarly the other two integrals can be shown to be equal)

$$I = \oint_C (F_1, 0)\cdot d\underline{s} = \int_a^b \underbrace{\underbrace{\underbrace{(F_1(x,\varphi(x)),0)}_{\underline{F}(\underline{c})}\cdot\underbrace{(1,\varphi'(x))}_{\underline{c}'(x)}}_{=F_1(x,\varphi(x))} dx}_{C_1}$$

$$- \int_a^b F_1(x,\psi(x))\,dx + \int_{C_2} \underbrace{(F_1,0)\cdot(0,*)}_{=0}\,dy$$

$$+ \int_{C_4} \underbrace{(F_1,0)\cdot(0,*)}_{=0}\,dy$$

$$= \int_a^b F_1(x,\psi(x)) - F_1(x,\varphi(x))\,dx$$



Only considering this special case !

$C_1: (x,\varphi(x)) = \underline{c}_1(x)$

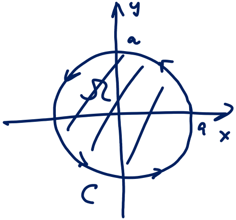
$$J = -\iint_\Omega \frac{\partial F_1}{\partial y}\,dx\,dy = -\int_a^b \int_{\varphi(x)}^{\psi(x)} \frac{\partial F_1}{\partial y}\,dy\,dx \underset{\uparrow}{=} \int_a^b -F_1(x,\psi(x)) + F_1(x,\varphi(x))\,dx$$

FTC

$\Rightarrow J = I$! $\square$

Example 14.14

Let $\mathbf{F} = -x^2y\mathbf{i} + xy^2\mathbf{j}$ and let the region Ω be given by $x^2 + y^2 \leq a^2$. Evaluate the line integral of $\mathbf{F}$ along the boundary of Ω.



Using Green's theorem:

$$\oint_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \iint_\Omega \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA \boxed{= \iint_\Omega \underbrace{y^2 + x^2}_{r^2} \, dA} =$$

$$F = \begin{pmatrix} -x^2y \\ xy^2 \end{pmatrix} \Rightarrow \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = y^2 + x^2$$

$$= \int_0^{2\pi} d\theta \int_0^a \underbrace{r^2 \cdot r}_{r^3} \, dr = 2\pi \cdot \frac{1}{4} a^4 = \frac{\pi a^4}{2}$$

Example 14.15

Compute the line integral

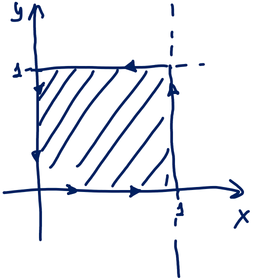
$$I = \oint_C \overset{F_2\,dy + F_1\,dx}{xy\,dy - y^2\,dx}, = \oint \underset{\sim}{F} \cdot d\underset{\sim}{s}$$

where the curve C is the boundary of the square cut out by the lines $x = 1$, $y = 1$ and the first quadrant, with anticlockwise orientation.

$$\underset{\sim}{F} = \begin{pmatrix} -y^2 \\ xy \end{pmatrix}$$

$$I \overset{\text{Green's}}{=} \int_0^1\int_0^1 (y + 2y)\,dx\,dy = 3\int_0^1 y\,dy$$

$$= 3 \cdot \frac{1}{2} = \frac{3}{2}$$



If we choose $\mathbf{F} = (F_1, F_2)$ in such a way that $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 1$, then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_\Omega \underbrace{\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)}_{1} dA = \text{area}(\Omega).$$

For instance, we can let $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = \frac{1}{2} + \frac{1}{2} = 1$

$$\mathbf{F} = -\frac{1}{2} y\mathbf{i} + \frac{1}{2} x\mathbf{j}, \quad \underline{\mathbf{F} = x\mathbf{j}}, \quad \underline{\mathbf{F} = -y\mathbf{i}}.$$

Then

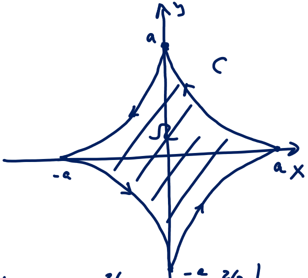
$$\frac{1}{2} \oint_C -ydx + xdy = \oint_C xdy = -\oint_C ydx = \text{area}(\Omega).$$

Example 14.16

Calculate the area bounded by the astroid

$$(x^2)^{1/3} + (y^2)^{1/3} = a^{2/3}, \quad a > 0,$$

using Green's theorem.



because of the choice of $\underset{\sim}{F}$

$$I = \oint_C \underset{\sim}{F} \cdot d\underset{\sim}{s} = \iint_\Omega 1 \, dx\,dy = \text{Area}(\Omega)$$

$$\underset{\sim}{c}(t) = \begin{pmatrix} a\cos^3 t \\ a\sin^3 t \end{pmatrix}$$

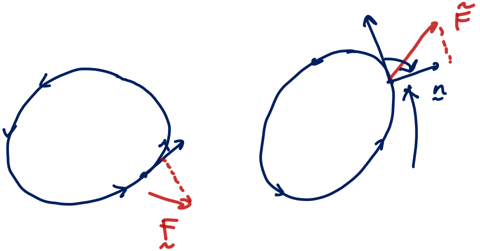
substitute:

$$(x^2)^{1/3} + (y^2)^{1/3} = a^{2/3}\cos^2 t + a^{2/3}\sin^2 t = a^{2/3}$$

choose $\underset{\sim}{F} = \begin{pmatrix} 0 \\ x \end{pmatrix}$

$$I = \int_0^{2\pi} \begin{pmatrix} 0 \\ a\cos^3 t \end{pmatrix} \cdot \begin{pmatrix} c_1'(t) \\ 3a\sin^2 t \cos t \end{pmatrix} dt = \int_0^{2\pi} 3a^2 \sin^2 t \cos^4 t \, dt = \ldots = \frac{3\pi a^2}{8}$$

Flux of a vector field



Flux integral:

$$\oint_C \underset{\sim}{F}\cdot\hat{\underset{\sim}{n}}\, ds = \int_a^b \underset{\sim}{F}\cdot\hat{\underset{\sim}{n}}\cdot \|\underset{\sim}{c}'(t)\|\, dt =$$

$$\hat{\underset{\sim}{n}} = \begin{pmatrix} c_2'(t) \\ -c_1'(t) \end{pmatrix} \Big/ \|\underset{\sim}{c}'(t)\|$$

$$= \int_a^b \underset{\sim}{F} \cdot \begin{pmatrix} c_2'(t) \\ -c_1'(t) \end{pmatrix} dt$$

$$= \int_a^b \left(F_1 \cdot c_2'(t) - F_2\, c_1'(t)\right) dt$$

$$G := (-F_2, F_1)$$

$$= \int_a^b G_2 \cdot c_2'(t) + G_1 \cdot c_1'(t)\, dt = \oint_C \underset{\sim}{G}\cdot d\underset{\sim}{s} = I$$

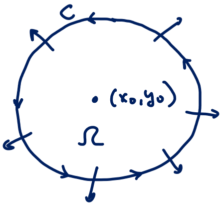
continued

$$I \underset{\text{Green's thm}}{=} \iint_\Omega \left(\frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y}\right) dx\,dy = \iint_\Omega \underbrace{\left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}\right)}_{\operatorname{div} \underset{\sim}{F}} dx\,dy$$

Green's theorem in normal form
(under the same assumptions as thm 13.14)

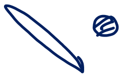
$$\oint_C \underset{\sim}{F}\cdot\hat{\underset{\sim}{n}}\, ds = \iint_\Omega \operatorname{div} \underset{\sim}{F}\, dx\,dy$$

$$\operatorname{div} \underset{\sim}{F} = \underbrace{\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}}\ (*)$$



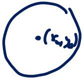
Define divergence:

$$\operatorname{div} \underset{\sim}{F} \overset{*}{=} \lim_{\operatorname{diam}(\Omega) \to 0} \frac{1}{\operatorname{Vol}(\Omega)} \oint_C \underset{\sim}{F} \cdot \underset{\sim}{\hat{n}}\, ds$$



$$= \lim_{\operatorname{diam}(\Omega) \to 0} \frac{1}{\operatorname{Vol}(\Omega)} \iint_\Omega \underbrace{\operatorname{div} \underset{\sim}{F}(x,y)}_{(*)\ \text{old definition}}\, dx\,dy = \boxed{\operatorname{div} \underset{\sim}{F}(x_0, y_0)}$$

Green's thm in normal form



$$\underbrace{\operatorname{div} \underset{\sim}{F}(x_0,y_0) - \varepsilon} \le \operatorname{div} \underset{\sim}{F}(x,y) \le \underbrace{\operatorname{div} \underset{\sim}{F}(x_0,y_0) + \varepsilon}$$

$\operatorname{div} \underset{\sim}{F}(x,y)$ is continuous $\Rightarrow \forall\, \varepsilon > 0\ \exists\, \delta > 0$ s.t. for all $(x,y) \in \underline{B}((x_0,y_0), \delta)$

we h

$$\underbrace{\frac{1}{A}\iint_{\Omega} \underbrace{\operatorname{div} \underset{\sim}{F}(x,y)}_{\leq}\, dx\, dy}_{L(\Omega)} \leq \frac{1}{A}\iint_{\Omega} (\underbrace{\overbrace{\operatorname{div} F(x_0,y_0)}^{L}+\varepsilon}) \, dx\, dy$$

$$= (\underbrace{\operatorname{di}\ldots}+\varepsilon) \cdot \cancel{\frac{1}{A}\iint_{\Omega} dx\, dy}$$

$$= \underbrace{\operatorname{div} F(x_0,y_0)}_{L} + \varepsilon$$

$$L-\varepsilon \leq L(\Omega) \leq L+\varepsilon$$

$$|L(\Omega)-L| \leq \varepsilon$$

To show that $\lim_{\operatorname{diam}(\Omega)\to 0} L(\Omega) = L$

we demonstrate that $\forall\, \varepsilon>0$ $\exists\, \delta>0$ s.t. for Ω: $\operatorname{diam}(\Omega)<\delta$

then $|L(\Omega)-L| < \varepsilon$